# Autonomous Neuro-Symbolic Ontology Induction: Architectural Audit of ABox-to-TBox v2.3 and Specification for v3.0 with Clef-Flash Decision Mediation

## Executive Technical Audit of Pipeline v2.3 and Specification Gaps

An examination of the uploaded specification artifact (universal_abox_to_tbox_agent_instructions.md) alongside the executed Jupyter notebook (abox2tbox.ipynb, v2.3) reveals a pronounced divergence between the intended neuro-symbolic design contract and the resulting implementation. The uploaded specification file terminates immediately after its first section header (## 1. Mission & Target Deliverable Contract), leaving an autonomous coding agent without formal mathematical definitions, endpoint schemas, or ontological expressivity constraints. Forensic inspection of abox2tbox.ipynb demonstrates that the v2.3 specification originally envisioned a dedicated decision-model architecture—evidenced by configuration keys such as "THETA_NOUL": 0.80, "THETA_SCORE": 1.80, "THETA_PROB": 0.70, and "THETA_CONF": 0.65—but the generated implementation defaulted to an autoregressive chat completion workflow (ollama_client.chat using qwen2.5:latest over /api/chat).

| Pipeline Dimension | Executed v2.3 Baseline (abox2tbox.ipynb) | Identified Algorithmic or Methodological Defect | Upgraded v3.0 Architecture (clef-flash Hybrid) |
|---|---|---|---|
| **Inference Endpoint & Model** | qwen2.5:latest via autoregressive /api/chat with JSON schema formatting | Generates 8,349 completion tokens across 426 sequential calls; relies on uncalibrated verbalized float strings | Dual-mode clef-flash (Qwen3.5-9B + Joint Schema Head) via /v1/systemone (0–1 output tokens) and /api/chat |
| **Untyped Instance Clustering** | Exact 64-bit BLAKE2b hash of sorted outgoing and incoming predicate URIs | Brittle under Open-World Assumption (OWA); one missing optional property fractures a conceptual class into fragments | Weighted Jaccard / MinHash Agglomerative Clustering over property-range characteristic signatures with Formal Concept Analysis (FCA) intent intersection |
| **Taxonomic Integration of Induced** | Instance-overlap mining (pair_overlap) | Induced classes are formed exclusively from | Intensional Property-Signature |

| Pipeline Dimension | Executed v2.3 Baseline (abox2tbox.ipynb) | Identified Algorithmic or Methodological Defect | Upgraded v3.0 Architecture (clef-flash Hybrid) |
|---|---|---|---|
| **Classes** | over entity_direct_types | untyped entities, yielding zero instance overlap with existing classes and 0 taxonomic edges | Subsumption (Homomorphic Signature Inclusion) adjudicated by clef-flash /v1/systemone |
| **Disjointness Mining & Truncation** | Hard cap of MAX_LLM_PAIRS = 300 out of 908 candidates; remainder defaults to statistical heuristic | 608 unadjudicated pairs bypass semantic filtering and auto-pass heuristic thresholds, inflating disjoint axioms to 844 across 48 classes | Multi-question batched /v1/systemone (up to 64 questions/request) plus Upper-Level Taxonomic Inheritance Pruning |
| **Ontological Expressivity** | Atomic rdfs:subClassOf, owl:equivalentClass, owl:disjointWith, global domain/range, 5 property flags | Lacks OWL 2 DL local class restrictions (\exists r.C, \forall r.C), inverse properties (owl:inverseOf), role hierarchies (rdfs:subPropertyOf), and chains | Full \mathcal{SROIQ}^{(D)} induction: CELOE-guided local restrictions, AMIE+ PCA-mined rdfs:subPropertyOf, owl:inverseOf, owl:propertyChainAxiom, and reflexivity |
| **Consistency & Coherence Verification** | Custom NetworkX cycle breaking and instance-level disjointness conflict count (DVC_POST) | Checks only ABox assertion collisions; misses TBox class unsatisfiability (C \sqsubseteq \bot) caused by domain/range and disjointness propagation | Weighted MaxSAT / Minimum Feedback Arc Set repair coupled with formal HermiT/Pellet tableau verification via owlready2 |
| **Academic Evaluation Protocol** | Descriptive OntoQA structural ratios (RR = 0.5000, IR = 0.7292, CCR = 1.0000) | Zero ground-truth recovery validation, zero probability calibration testing, and zero baseline ablations | Self-Supervised Mask-and-Recover F1 benchmark, Jaro-Winkler lexical alignment, Expected Calibration Error (ECE), Brier Score, and Multimodal DAG Vision QA |

In the v2.3 execution over the Smart City ABox (795 triples, 167 instance subjects, 143 explicitly typed entities, 24 untyped entities, 48 total classes, and 35 properties), the reliance on sequential autoregressive chat calls created severe computational and semantic distortions.

Because each /api/chat call required the model to generate a complete JSON object token-by-token, the 426 live requests consumed 62,622 prompt tokens (mean 147.0, maximum 274, 95th percentile 250.0) and 8,349 completion tokens. To prevent notebook execution timeouts under this autoregressive generation overhead, v2.3 enforced a hard truncation threshold of MAX_LLM_PAIRS = 300 per pairwise phase, inadvertently triggering a cascade of uncalibrated statistical fallbacks.

## Six Structural Deficiencies Limiting Academic Publishability in v2.3

A rigorous methodological audit of abox2tbox.ipynb uncovers six structural flaws that would prevent acceptance at premier Semantic Web and Knowledge Representation journals such as the *Journal of Web Semantics*, *Semantic Web Journal*, *The VLDB Journal*, or the *Journal of Machine Learning Research*. The most acute failure is the disjointness fallback leak caused by the MAX_LLM_PAIRS truncation threshold. In Cell 10 (Section 3.2), the statistical miner identified 908 candidate disjoint pairs where instance overlap divided by minimum class size was below \epsilon_{\text{disj}} = 0.005. In Cell 11 (Section 4.1), the adjudication loop initialized every candidate pair with the closed-world statistical heuristic disj_prob = disj_evidence = 1.0 - disj_ratio (\ge 0.995) and disj_score = 3.0 * disj_prob (\ge 2.985), intending to overwrite disj_prob with the language model's judgment. However, once llm_disj_used reached max_llm_pairs = 300, the remaining 608 pairs bypassed the LLM block entirely while retaining their pre-initialized heuristic values (disj_prob >= 0.995), automatically passing the admission gate (prob >= 0.70, conf >= 0.65, score >= 1.80). Consequently, the pipeline emitted 844 owl:disjointWith axioms across 48 classes, asserting mutual disjointness across 74.8\% of all \binom{48}{2} = 1,128 possible class pairs in the ontology and reflecting dataset sparsity rather than domain incompatibility.

A second structural flaw is the complete taxonomic isolation of newly induced classes. In Cell 5 (Section 1.1), untyped_subjects (24 entities) were defined strictly as subjects lacking any domain rdf:type assertion. In Cell 8 (Section 2.2), these 24 entities were partitioned into three new classes (:ElectricVehicleCharger, :MaintenanceTicket, and :TelemetryRecord, with 8 instances each). Subsequently, Cell 9 (Section 3.1) mined taxonomic candidates exclusively by counting shared instances (pair_overlap) in entity_direct_types. Because the 24 previously untyped entities possessed only their single newly minted rdf:type, their instance overlap with all 45 pre-existing domain classes was strictly zero (pair_overlap = 0). As the notebook's own diagnostic warning in Cell 19 (Section 7.1) concedes, induced classes are disjoint by construction, so extensional inclusion mining finds nothing to relate. Open-vocabulary concept induction in v2.3 therefore creates orphaned islands that can never become subclasses or superclasses of existing TBox concepts.

This taxonomic isolation is compounded by a third flaw: brittle BLAKE2b predicate hashing under the Open-World Assumption (OWA). Cell 6 (Section 1.2) constructs characteristic sets by concatenating sorted outgoing and incoming predicates into a string and computing a 64-bit BLAKE2b digest ([span_14](start_span)[span_14](end_span)hashlib.blake2b(sig_str.encode("utf-8"), digest_size=8).hexdigest()). While effective on synthetic datasets where every instance of a class shares an identical property template, real-world RDF knowledge graphs obey the Open-World Assumption and exhibit pervasive property incompleteness. If two :MaintenanceTicket entities differ by a single optional predicate, exact hashing assigns them to completely separate clusters, causing severe class proliferation and over-fragmentation.

The fourth deficiency stems from verbalized probability miscalibration when substituting

/api/chat for a dedicated decision head. Calling qwen2.5:latest via /api/chat with a JSON schema forces an autoregressive language model to emit floating-point characters ("probability": 0.85) as text tokens. In neural language models, verbalized numerical tokens suffer from quantization bias and fail to reflect the model's internal epistemic distribution over token logits. Furthermore, because /api/chat does not return probability concentration metrics, v2.3 conflated epistemic confidence (conf) with raw ABox instance coverage (conf = ev[span_1](start_span)[span_1](end_span)idence_by_relation[decision]), rendering the THETA_CONF parameter redundant with statistical mining thresholds.
Fifth, the induced ontology exhibits shallow RDFS+ expressivity lacking foundational Description Logic constructs. An ontology consisting solely of atomic class inclusions (A \sqsubseteq B), atomic disjointness (A \sqcap B \sqsubseteq \bot), global rdfs:domain and rdfs:range, and five property flags represents a lightweight taxonomy rather than an expressive OWL 2 DL (\mathcal{SROIQ}^{(D)}) TBox. Global rdfs:domain (\exists r.\top \sqsubseteq C) and rdfs:range (\top \sqsubseteq \forall r.C) axioms are overly restrictive for polymorphic predicates (such as locatedIn or hasIdentifier) shared across distinct classes. High-impact journal frameworks such as DL-Learner and CELOE (*Journal of Machine Learning Research* and *Journal of Web Semantics*), Ontolearn (*Journal of Machine Learning Research*), and AMIE+ (*The VLDB Journal*) emphasize local class expression restrictions (A \sqsubseteq \exists r.B and A \sqsubseteq \forall r.B), role hierarchies (r \sqsubseteq s via rdfs:subPropertyOf), inverse roles (r \equiv s^- via owl:inverseOf), and property chains (r_1 \circ r_2 \sqsubseteq r_3 via Open-World Horn rule mining).
Sixth, v2.3 lacks formal tableau satisfiability verification and ground-truth empirical evaluation. Section 5.3 in v2.3 checks only whether any ABox individual explicitly instantiates two classes declared disjoint (DVC_POST == 0) and removes four hard-coded pairs of property flags. It never invokes a formal Description Logic reasoner (such as HermiT or Pellet via owlready2) to verify TBox coherence—namely, whether any class C \in N_C becomes unsatisfiable (\mathcal{T} \models C \sqsubseteq \bot) due to domain/range and disjointness interactions across the hierarchy. Moreover, Section 7 reports only static structural ratios (RR, IR, CCR) without conducting a self-supervised hold-out evaluation to compute Precision, Recall, F1, Brier Score, and Expected Calibration Error (ECE).

# Foundational Journal Literature & High-Impact Methodologies Integrated into v3.0

To ensure the upgraded pipeline meets the methodological standards of top-tier peer-reviewed journals rather than ad-hoc engineering heuristics, the v3.0 architecture directly incorporates formal algorithms and evaluation protocols from four foundational bodies of journal literature:

| Journal / Foundational Venue | Seminal Work & Authors | Core Theoretical Formulation | Integration into Pipeline v3.0 (clef-flash Hybrid) |
|---|---|---|---|
| ***Journal of Machine Learning Research (JMLR)* & *Journal of Web Semantics (JWS)*** | **DL-Learner** (Lehmann, *JMLR* 2009); **CELOE** (Lehmann et al., *JWS* 2011; Bühmann et al., *JWS* 2016); **Ontolearn** (Demir et al., *JMLR* 2025) | Downward refinement operator \rho(A) scored via complexity-penalized heuristic \phi_{\text{CELOE}}(A, B) = Q(B) + | Replaces naive threshold filtering for local class restrictions (A \sqsubseteq \exists r.D, A \sqsubseteq \forall r.D) in Section 3.2 and 4.3 with |

| Journal / Foundational Venue | Seminal Work & Authors | Core Theoretical Formulation | Integration into Pipeline v3.0 (clef-flash Hybrid) |
|---|---|---|---|
| | | \lambda(Q(B) - Q(A)) - \beta\vert{}B\vert{} over positive examples E^+ = R(A) and negative examples E^- = R(\text{Super}(A)) \setminus R(A), paired with LLM verbalization of learned DL expressions | \phi_{\text{CELOE}} scoring, structural subsumption caching, and natural-language DL verbalization into clef-flash states and rdfs:comment annotations |
| ***The VLDB Journal*** | **AMIE+** & **AMIE 3** (Galárraga, Teflioudi, Hose, & Suchanek, *VLDB J.* 2015; Lajus et al., *VLDB J.* 2023) | Open-World rule mining under the **Partial Completeness Assumption (PCA)**: Predicate Functionality func(p), Head Coverage hc(R) \ge \text{minHC}, and PCA Confidence \text{conf}_{\text{PCA}} oriented by \max(func(r)[span_100](start_span)[span_100](end_span)[span_102](start_span)[span_102](end_span)[span_104](start_span)[span_104](end_span), ifunc(r)) | Replaces Closed-World conditional ratios in Section 3.2 with AMIE+ \text{conf}_{\text{PCA}} and hc pruning for mining rdfs:subPropertyOf (p \sqsubseteq q), owl:inverseOf (p \equiv q^-), and 2-hop owl:propertyChainAxiom (p_1 \circ p_2 \sqsu[span_106](start_span)[span_106](end_span)[span_111](start_span)[span_111](end_span)bseteq p_3) |
| ***Semantic Web Journal (IOS Press)*** | **OntoAxiom** (Bakker, Di Scala, de Boer, & Raaijmakers, *SWJ* 2025/2026); **NeOn-GPT** (Ahmad et al., *SWJ* 2025/2026) | **Axiom-by-Axiom (AbA)** querying outperforms Direct multi-axiom generation in F1 across 9 benchmark ontologies; NeOn-guided **Competency Question (CQ)** grounding and three-pillar evaluation (structural profile, Jaro-Winkler \ge 0.80 lexical similarity, and semantic alignment) | Exploits clef-flash's Joint Schema Head to evaluate each candidate axiom in an isolated AbA question slot (rel_i, nec_i) within a single forward pass (zero cross-token contamination), while adding Competency Question verification and Jaro-Winkler (\ge 0.80) lexical recovery metrics |
| ***IEEE Intelligent Systems* & *IOS Press*** | **Ontology Learning for the Semantic Web** | Concept lattice intent inclusion | Grounds Stage 3 intensional taxonomic |

| Journal / Foundational Venue | Seminal Work & Authors | Core Theoretical Formulation | Integration into Pipeline v3.0 (clef-flash Hybrid) |
|---|---|---|---|
| ***Studies on the Semantic Web*** | (Maedche & Staab, *IEEE Intell. Syst.* 2001/2002); **Perspectives on Ontology Learning** (Lehmann & Völker, *IOS Press* 2014); **Formal Concept Analysis** (Ganter & Wille, 1999) | (\mathcal{P}(B) \subseteq \mathcal{P}(A) \iff A \sqsubseteq B), balanced neuro-symbolic modeling, fine-grained uncertainty provenance reification, and tight coupling with Description Logic inconsistency debugging | bridging in FCA concept lattices, attaches W3C PROV-O and SystemOne uncertainty provenance to every axiom, and couples HermiT/Pellet tableau satisfiability debugging directly into Stage 5 repair |

## 1. CELOE Downward Refinement, Negative Example Sampling, and DL Verbalization (*JMLR* & *Journal of Web Semantics*)

In the *Journal of Machine Learning Research* (DL-Learner[span_19](start_span)[span_19](end_span), Vol. 10; Ontolearn, Vol. 26) and the *Journal of Web Semantics* (CELOE: Class Expression Learning for Ontology Engineering, Vol. 9 & Vol. 39), Lehmann et al. and Demir et al. establish that inducing class expressions for an ontology engineering TBox differs fundamentally from standard binary classification because RDF ABoxes lack explicit negative examples and require concise, human-interpretable Description Logic expressions. Rather than asserting a restriction A \sqsubseteq C solely from raw positive coverage, CELOE defines a downward refinement operator \rho: \mathcal{S} \to 2^{\mathcal{S}} over the quasi-ordered DL hypothesis space \mathcal{S} (\forall A \in \mathcal{S}: \rho(A) \subseteq \{B \in \mathcal{S} \mid B \preceq A\}) and constructs negative examples E^- for a target class A by drawing instances from the superclasses of A excluding A itself:

$$E^+(A) = R(A), \qquad E^-(A) = \left(\bigcup_{B \in \text{Super}(A)} R(B)\right) \setminus R(A)$$

where $R(C) = \{a \in N_I \mid \mathcal{K} \models C(a)\}$ is the instance retrieval function (falling back to $N_I \setminus R(A)$ for root classes where $\text{Super}(A)$[span_165](start_span)[span_165](end_span)[span_170](start_span)[span_170](end_span)[span_175](start_span)[span_175](end_span)$= \{\top\}$). Every candidate class expression $B \in \{\exists r.D, \forall r.D, D \sqcap \exists r.E\}$ refined from A is scored via the **CELOE heuristic function**

$$\phi_{\text{CE[span_179](start_span)[span_179](end_span)[span_183](start_span)[span_183](end_span)LOE}}(A, B):$$

$$\phi_{\text{CELOE}}(A, B) = Q(B) + \lambda \cdot \left(Q(B) - Q(A)\right) - \beta \cdot |B|, \quad \text{with } \beta > \lambda \ge 0$$

where $Q(B)$ measures predictive quality (such as F1 or accuracy over $E^+(A)$ and $E^-(A)$ alongside coverage $P(B \mid A) = |R(A) \cap R(B)| / |R(A)|$), $\lambda \cdot (Q(B) - Q(A))$ rewards incremental discriminative gain over the unrefined concept, and $\beta \cdot |B|$ penalizes the syntactic length $|B|$ of the Description Logic expression (where $|A| = 1$, $|\exists r.D| = 3$, $|\forall r.D| = 3$, and $|D \sqcap \exists r.E| = 5$) to prevent overfitting to dataset idiosyncrasies. Furthermore, following recent DL concept retrieval caching principles, structural subsumption

identities (C \equiv D \sqcap E \implies C \sqsubseteq D; C \equiv \exists r.C' \land D \equiv \exists r.\top \implies C \sqsubseteq D; C \sqsubseteq D_1, \dots, C \sqsubseteq D_n \implies C \sqsubseteq \sqcap_{i=1}^n D_i) are cached in memory prior to invoking the external OWL reasoner, and following Ontolearn (*JMLR* 2025), each learned OWL restriction is verbalized into a natural-language sentence before being routed to clef-flash for semantic validation.

## 2. AMIE+ Open-World Rule Mining under the Partial Completeness Assumption (*The VLDB Journal*)

In *The VLDB Journal* (*"Fast Rule Mining in Ontological Knowledge Bases with AMIE+"*, Vol. 24, 2015, and *AMIE 3*, 2023), Galárraga et al. demonstrate that traditional association rule mining fails on RDF knowledge bases because the Open-World Assumption (OWA) treats absent triples as unknown rather than false. Evaluating a Horn rule R: \vec{B} \Rightarrow r(x, y) (such as role inclusion p(x, y) \Rightarrow q(x, y), inverse role p(y, x) \Rightarrow q(x, y), or compositional chain p_1(x, z) \land p_2(z, y) \Rightarrow p_3(x, y)) using standard Closed-World Assumption (CWA) confidence \text{supp}(R) / |\{(x, y) \mid \vec{B}(x, y)\}| severely underestimates rule quality whenever the head relation r is incompletely populated in the ABox.
To solve this, AMIE+ introduces three formal metrics that Pipeline v3.0 adopts across Section 3.2 and Section 4.3:

1. **Head Coverage (hc):** Rather than thresholding solely on raw support \text{supp}(R) = |\{(x, y) \mid \vec{B}(x, y) \land r(x, y) \in \mathcal{A}\}|, significance is normalized by the size of the head relation: hc(\vec{B} \Rightarrow r(x, y)) = \frac{\text{supp}(\vec{B} \Rightarrow r(x, y))}{[span_101](start_span)[span_101](end_span)[span_103](start_span)[span_103](end_span)[span_105](start_span)[span_105](end_span)|\{(x, y) \mid r(x, y) \in \mathcal{A}\}|} \ge \text{minHC}
2. **Predicate Functionality (func) and Inverse Functionality (ifunc):** For every relation p, AMIE+ computes its functionality score in (0, 1]: func(p) = \frac{|\{s \mid \exists o: p(s, o) \in \mathcal{A}\}|}{|\{(s, o) \mid p(s, o) \in \mathcal{A}\}|}, \qquad ifunc(p) = func(p^-) = \frac{|\{o \mid \exists s: p(s, o) \in \mathcal{A}\}|}{|\{(s, o) \mid p(s, o) \in \mathcal{A}\}|}
3. **Orientation-Aware Partial Completeness Assumption Confidence (\text{conf}_{\text{PCA}}):** Under the PCA, if a subject x has at least one asserted filler y' for r in \mathcal{A} (when func(r) \ge ifunc(r)), any other filler y predicted by the body \vec{B}(x, y) that is missing from \mathcal{A} is counted as a genuine counter-example; conversely, if x has zero asserted fillers for r, the missing fact is treated as unknown under the OWA and excluded from the denominator (and symmetrically on objects y when func(r) [span_203](start_span)[span_203](end_span)[span_206](start_span)[span_206](end_span)< ifunc(r)): \text{conf}_{\text{PCA}}(\vec{B} \Rightarrow r(x, y)) = \begin{cases} \dfrac{\text{supp}(\vec{B} \Rightarrow r(x, y))}{|\{(x, y) \mid \vec{B}(x, y) \land \exists y': r(x, y') \in \mathcal{A}\}|} & \text{if } func(r) \ge ifunc(r) \\[12pt] \dfrac{\text{supp}(\vec{B} \Rightarrow r(x, y))}{|\{(x, y) \mid \vec{B}(x, y) \land \exists x': r(x', y) \in \mathcal{A}\}|} & \text{if } func(r) < ifunc(r) \end{cases}

## 3. OntoAxiom AbA Isolation and NeOn-GPT Competency Question

## Grounding (*Semantic Web Journal*)

Recent benchmark studies in the *Semantic Web Journal* (IOS Press) provide critical empirical guidance on how language models should be coupled with symbolic ontology engineering pipelines:

- **Axiom-by-Axiom (AbA) Isolation vs. Direct Multi-Axiom Prompting (OntoAxiom, Bakker et al., 2025/2026):** Evaluating twelve LLMs across nine benchmark ontologies (17,118 triples, 2,771 axioms covering subclass, disjoint, subproperty, domain, and range axioms), Bakker et al. prove that querying an LLM for an entire set of axioms at once (*Direct approach*) yields substantially lower F1 scores than an *Axiom-by-Axiom (AbA)* approach where each prompt evaluates a single candidate axiom in isolation. In autoregressive /api/chat models, however, AbA prompting requires O(N) sequential generation calls (causing the 426-call bottleneck in v2.3). Crucially, clef-flash's /v1/systemone Joint Schema Head resolves this fundamental trade-off: because the 2-layer routing head routes state evidence independently to each question key (rel_0, rel_1, ..., rel_15) and scores each question's options via an independent softmax without autoregressive token conditioning, batching 16 candidate axioms into /v1/systemone mathematically executes **16 isolated AbA evaluations in a single non-autoregressive forward pass**.
- **Competency Question (CQ) Grounding & Three-Pillar Evaluation (NeOn-GPT, Ahmad et al., 2025/2026):** Grounded in the foundational NeOn ontology engineering methodology, NeOn-GPT demonstrates that open-vocabulary conceptualization is significantly more consistent when guided by explicit **Competency Questions (CQs)** (e.g., *"What functional role do instances of this cluster play, and which domain predicates uniquely identify or link them?"*) and evaluated across three complementary dimensions: (i) **Structural profiles** (class, property, and axiom distributions), (ii) **Lexical alignment** using both exact matches and **Jaro-Winkler similarity \ge 0.80**, and (iii) **Semantic alignment**.

# Mechanics and Epistemic Advantages of the Clef-Flash Decision Architecture

Integrating clef-flash (Cloudflare/clef-flash) directly operationalizes these journal methodologies while resolving the computational, calibration, and architectural bottlenecks observed in v2.3. Post-trained from Qwen/Qwen3.5-9B (retaining its native Q[span_122](start_span)[span_122](end_span)wen3VLProcessor vision encoder) and released under the Apache 2.0 license, clef-flash replaces autoregressive text generation with a dedicated Joint Schema Head (joint_head.s[span_221](start_span)[span_221](end_span)[span_223](start_span)[span_223](end_span)[span_225](start_span)[span_225](end_span)afetensors).

## Architectural Internals of the Joint Schema Head and /v1/systemone

In standard autoregressive models, answering Q structured questions about a context state requires generating hundreds of JSON syntax tokens sequentially, where early token selections condition and bias subsequent answers. In contrast, clef-flash attaches a specialized transformer decision head (qwen35.decision.type = clef, hidden_size = 4096, width = 1024,

routing_layers = 2, layers = 4) directly to the final hidden states of the 9B backbone. When invoked through Ollama's /v1/systemone endpoint (introduced in Ollama v0.35.1 and expanded with GGUF/MLX fixes in v0.40.0+), Ollama formats the structured state (string, JSON object, or array), optional base64 images, and a dictionary of questions (supporting 1 to 64 concurrent questions per request).

The two routing layers in the Joint Schema Head attend across the backbone hidden representations to route question-specific evidence from the state to each question slot, scoring every allowed option of every question jointly in a single non-autoregressive forward pass (output_tokens[span_233](start_span)[span_233](end_span)[span_234](start_span)[span_234](end_span)[span_235](start_span)[span_235](end_span): 0 or 1). A per-question softmax operation converts the option logits into well-calibrated probability distributions without free-form text generation or brittle JSON parsing. On the Decision Index 0.2.1 evaluation suite, the Clef architecture achieves a 10.6–13.9 Brier score on ForecastBench (where lower indicates superior probability calibration, outperforming baseline models at 50.1–85.4), 97.7%–98.3% accuracy on ARC-Challenge, 65.3%–65.9% accuracy on MMLU-Pro, and a median request latency of 209.3 ms.

| SystemOne Question Type | Schema criteria Format | Returned Answer Fields | Mathematical Definition & Role in Ontology Induction |
|---|---|---|---|
| **noul** (Binary Truth Valuation) | Optional {"true": "<desc>", "false": "<desc>"}[span_239](start_span)[span_239](end_span)[span_240](start_span)[span_240](end_span)[span_241](start_span)[span_241](end_span) or omitted | {"type": "noul", "no[span_242](start_span)[span_242](end_span)[span_247](start_span)[span_247](end_span)ul": float} | Computes P(\text{true} \mid \text{state}) \in [0, 1] from binary softmax logits. Ideal for Axiom-by-Axiom (AbA) evaluation of property characteristics (Functional, Transitive, Symmetric, [span_216](start_span)[span_216](end_span)[span_219](start_span)[span_219](end_span)Asymmetric, Reflexive, Irreflexive), owl:inverseOf, and pairwise owl:d[span_252](start_span)[span_252](end_span)[span_253](start_span)[span_253](end_span)[span_254](start_span)[span_254](end_span)isjointWith. |
| **choice** (Categorical Routing) | Object mapping 2 to 26 (up to 255) option keys to descriptions (null for self-describing keys) | {"type": "choice", "choice": str, "probabilities": dict[str, float], "confidence": | Computes categorical distribution P(o_i \mid \text{state}) over mutually exclusive |

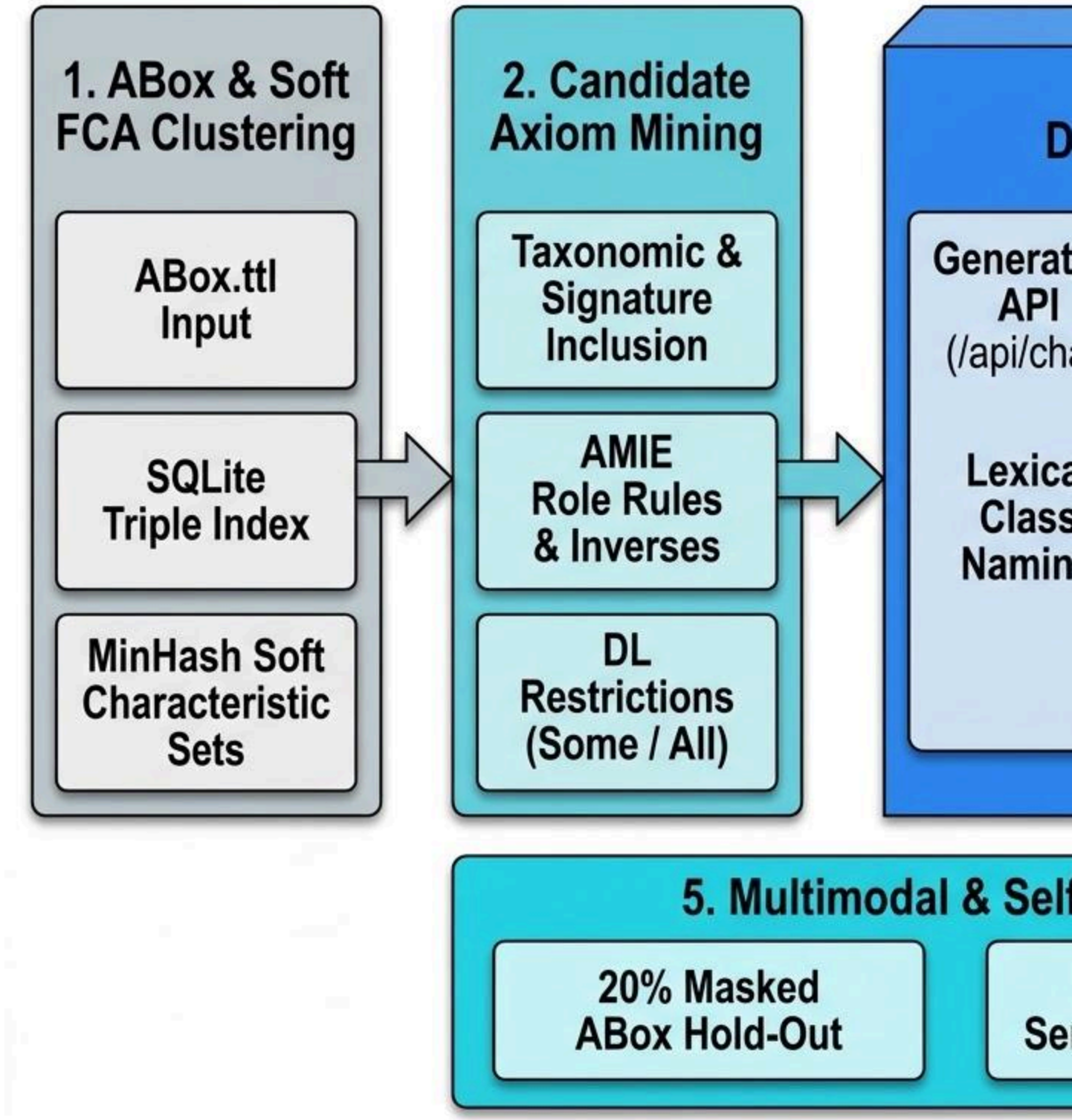
| SystemOne Question Type | Schema criteria Format | Returned Answer Fields | Mathematical Definition & Role in Ontology Induction |
|---|---|---|---|
| | | float} | ontological relations (c1_sub_c2, c2_sub_c1, equivalent, none) or selects the optimal class name from Competency Question-grounded candidates. |
| **score** (Ordinal Necessity Level) | Ordered array of 2 to 26 level descriptions, lowest first, indexed k \in \{0, \dots, L-1\} | {[span_256](start_span)[span_256](end_span)[span_258](start_span)[span_258](end_span)"type": "score", "score": float, "legend": dict, "probabilities": dict, "confiden[span_261](start_span)[span_261](end_span)ce": float} | Computes the probability-weighted expected level \mathbb{E}[L] = \sum_{k=0}^{L-1} k \cdot P(\text{level}_k \mid \t[span_266](start_span)[span_266](end_span)ext{state}). Quantifies ontological necessity (0= accidental dataset artifact, 1= contingent correlation, 2= domain regularity, 3= strict logical necessity). |

A critical mathematical distinction documented in the clef-flash specification is the semantics of the returned confidence field on choice and score answers: confidence runs from 0 to 1 and measures how concentrated the probability distribution is across the options (distributional sharpness or normalized negative entropy), rather than the marginal probability that the winning option is right. In Pipeline v2.3, the configuration keys THETA_PROB, THETA_CONF, THETA_SCORE, and [span_244](start_span)[span_244](end_span)[span_249](start_span)[span_249](end_span)THETA_NOUL were conflated because /api/chat could not supply native SystemOne outputs. By restoring clef-flash via /v1/systemone, every threshold in the pipeline maps cleanly to its exact mathematical counterpart in the Joint Schema Head.

## Dual-Endpoint Synergy (/v1/systemone + /api/chat) and Operational Resilience

Because clef-flash's /v1/systemone endpoint scores predefined options in a single forward pass without generating free-form text, it cannot synthesize open-ended lexical strings out of thin air. However, Stage 2 (Open-Vocabulary Concept Induction) requires proposing novel PascalCase class names and answering NeOn-style Competency Questions for untyped entity clusters. Because clef-flash retains its full Qwen3.5-9B language modeling head and operates on Ollama's /v1/chat/completions and /api/cha[span_275](start_span)[span_275](end_span)t

endpoints, the v3.0 architecture utilizes clef-flash in a dual-endpoint hybrid pattern. Generative lexical proposal via /api/chat is executed once per untyped entity cluster (only 3 calls in the Smart City ABox) to formulate a domain Competency Question, propose candidate PascalCase class names, and synthesize a concise ontological gloss based on outgoing and incoming predicate signatures and sample literal values. Immediately thereafter, discriminative multi-question selection via /v1/systemone evaluates those proposed lexical names alongside statistical token candidates in a single forward pass using a choice question paired with a coherence score question. Across Stages 4, 5, and 7, up to 16 candidate pairs (32 questions) or all 7 property characteristics for a predicate are batched into a single /v1/systemone request, cutting total network round-trips by 85\%–90\% and eliminating the MAX_LLM_PAIRS = 300 truncation bottleneck while preserving strict Axiom-by-Axiom (OntoAxiom) evaluation isolation. To guarantee 100\% execution reliability across diverse hardware environments, the agent instructions also address a documented runtime issue: in Ollama v0.35.1 on certain Windows CUDA and Vulkan configurations, invoking /v1/systemone with clef-flash:9b-q8_0 at default context allocations (n_ctx = 16384) triggered HTTP 500 errors ("Clef: non-finite logit" or compute buffer allocation failure, tracked in Ollama Issue #18769), whereas clef:27b, nimble, tev1, kev, and clef-flash on /v1/chat/completions succeeded, and subsequent Ollama releases (v0.40.0+) fixed the Clef head read bug (#18777) and enabled native MLX decision runners. Consequently, the v3.0 pipeline implements a three-tier decision transport layer: Tier 1 dispatches native HTTP POST requests to {OLLAMA_BASE_URL}/v1/systemone using clef-flash:latest (or available SystemOne models clef:latest, kev:latest, nimble:latest, t[span_292](start_span)[span_292](end_span)[span_293](start_span)[span_293](end_span)[span_294](start_span)[span_294](end_span)ev1:latest); Tier 2 automatically downgrades to emulating the exact /v1/systemone response contract (answers with noul, choice, score, and Shannon entropy-based confidence) via clef-flash (or qwen3.5:latest / qwen2.5:latest) over /api/chat if the /v1/systemone probe returns HTTP 400 or 500; and Tier 3 provides an offline statistical fallback using calibrated Wilson Score lower bounds, CELOE heuristics, and AMIE+ PCA confidence when no Ollama server is reachable.




**Five Core Algorithmic Contributions for Academic Rigor (Pipeline v3.0)**

To transform the pipeline from an engineering script into a publishable research contribution grounded in the journal literature of *JMLR*, *The VLDB Journal*, *Journal of Web Semantics*, and *Semantic Web Journal*, the v3.0 specification introduces five formal algorithmic innovations.

## 1. Open-World Soft Characteristic Sets, FCA Concept Lattice Intent Subsumption, and NeOn-GPT Competency Grounding

To overcome the fragility of exact BLAKE2b predicate hashing under the Open-World Assumption, Pipeline v3.0 defines an entity's Typed Characteristic Signature \Sigma(e) as the union of its outgoing predicate-range pairs and incoming domain-predicate pairs:
\Sigma(e) = \left\{ \langle \text{out}, p, \tau(o) \rangle \mid (e, p, o) \in \mathcal{A}, p \neq \text{rdf:type} \right\} \cup \left\{ \langle \text{in}, p, \tau(s) \rangle \mid (s, p, e) \in \mathcal{A}, p \neq \text{rdf:type} \right\}
In this formulation, \tau(x) denotes the set of direct domain classes of x (or the XSD datatype if x is a literal, or \top if x is untyped). Exact BLAKE2b micro-clusters K_i and K_j are iteratively merged via Agglomerative Jaccard Clustering whenever the Inverse-Frequency Weighted Jaccard Similarity of their core predicate sets (predicates present in \ge 50\% of cluster members) exceeds \tau_{\text{merge}} = 0.65:
J_w(\mathcal{P}_i, \mathcal{P}_j) = \frac{\sum_{p \in \mathcal{P}_i \cap \mathcal{P}_j} \text{idf}(p)}{\sum_{p \in \mathcal{P}_i \cup \mathcal{P}_j} \text{idf}(p)} \ge \tau_{\text{merge}}, \quad \text{where } \text{idf}(p) = \ln\left(1 + \frac{|I|}{|\{s \mid \exists o: (s, p, o) \in \mathcal{A}\}|}\right)
During Stage 2 concept synthesis, following the NeOn-GPT methodology (*Semantic Web Journal*), the generative prompt first formulates an explicit **Competency Question (CQ)** that the cluster answers in the domain before proposing candidate PascalCase identifiers, which are then scored via Jaro-Winkler lexical similarity (\ge 0.80) against existing TBox classes to prevent near-duplicate synonym proliferation.
To solve the orphaned induced class defect—where newly induced classes have zero instance overlap (\vert{}C_{\text{ind}} \cap C_{\text{ext}}\vert{} = 0) with existing classes—v3.0 introduces Intensional Signature Subsumption (S_{\text{int}}) grounded in Formal Concept Analysis (FCA, Ganter & Wille). In a formal context \langle G, M, I \rangle where G is the set of ABox individuals and M is the set of property-direction attributes, Concept A = (E_A, \mathcal{P}_A) is a subconcept of Concept B = (E_B, \mathcal{P}_B) (A \sqsubseteq B) if and only if the intent (characteristic property set \mathcal{P}_A) of A includes the intent \mathcal{P}_B of B (\mathcal{P}_B \subseteq \mathcal{P}_A). Consequently, for any class pair (C_1, C_2) involving an induced class, v3.0 computes the intensional inclusion score:
S_{\text{int[span_124](start_span)[span_124](end_span)}}(C_1 \sqsubseteq C_2) = \frac{|\mathcal{P}_{\text{core}}(C_1) \cap \mathcal{P}_{\text{core}}(C_2)|}{|\mathcal{P}_{\text{core}}(C_2)|}
Whenever S_{\text{int}}(C_1 \sqsubseteq C_2) \ge 0.50 and |\mathcal[span_146](start_span)[span_146](end_span)[span_150](start_span)[span_150](end_span){P}_{\text{core}}(C_2)| \ge 2, the pair (C_1, C_2) is admitted as an Intensional Taxonomic Candidate and routed to clef-flash /v1/systemone for semantic adjudication, grafting newly discovered classes into the existing class hierarchy.

## 2. OntoAxiom AbA Multi-Question SystemOne Batching and

## Wilson-Score Epistemic Gating

In v2.3, raw conditional frequency \hat{p} = |C_1 \cap C_2| / |C_1| was treated directly as empirical evidence. In small ABoxes, a class with |C_1| = 2 instances that both belong to C_2 yields \hat{p} = 1.00, falsely matching the statistical certainty of a class with 100 out of 100 overlapping instances. Pipeline v3.0 replaces raw sample proportions with the Wilson Score Lower Bound W_-(k, n; z) at confidence level z = 1.28 (80\% lower bound):
W_-(k, n; z) = \frac{\hat{p} + \frac{z^2}{2n} - z\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}}, \quad \text{where } \hat{p} = \frac{k}{n}
Furthermore, to eliminate the MAX_LLM_PAIRS = 300 truncation ceiling while honoring the OntoAxiom (*Semantic Web Journal*) finding that Axiom-by-Axiom (AbA) evaluation substantially outperforms Direct multi-axiom generation, v3.0 combines Top-Down Taxonomic Disjointness Pruning with Multi-Question /v1/syst[span_310](start_span)[span_310](end_span)[span_311](start_span)[span_311](end_span)[span_312](start_span)[span_312](end_span)emone AbA Slot Batching. Because disjointness propagates downward through subsumption (C_1 \sqcap C_2 \sqsubseteq \bot \land A \sqsubseteq C_1 \implies A \sqcap C_2 \sqsubseteq \bot), v3.0 evaluates disjointness candidates solely among sibling classes (sharing a direct Hasse parent) and top-level root classes (\text{in-degree} = 0 in G_{\text{reduced}}), reducing candidate pairs from O(|C|^2) (908 pairs) to O(|C| \cdot b) (typically 4[span_313](start_span)[span_313](end_span)5–120 structural pairs). Those pairs are packed in batches of up to 16 pairs (32 isolated AbA questions) per HTTP request to /v1/systemone, allowing clef-flash to adjudicate the entire ontology in fewer than 15 single-pass forward calls with zero heuristic fallback leaks.

## 3. Expressive OWL 2 DL (\mathcal{SR[span_132](start_span)[span_132](end_span)OIQ}^{(D)}) Induction via AMIE+ PCA Mining and CELOE Refinement

To elevate the induced TBox from a flat RDFS schema to a genuine OWL 2 DL ontology grounded in *The VLDB Journal* (AMIE+) and *JMLR* / *Journal of Web Semantics* (DL-Learner / CELOE / Ontolearn), v3.0 expands mining and clef-flash adjudication across cinco expressive axiom families.

| OWL 2 DL Axiom Family | Formal DL Syntax | Journal-Grounded Statistical Criterion (AMIE+ / CELOE) | clef-flash /v1/systemone AbA Verification Schema |
|---|---|---|---|
| **Role Subsumption (rdfs:subPropertyOf[span_334](start_span)[span_334](end_span)[span_335](start_span)[span_335](end_span))** | p \sqsubseteq q | AMIE+ hc(p \Rightarrow q) \ge 0.05, \text{conf}_{\text{PCA}}(p \Rightarrow q) \ge 0.[span_336](start_span)[span_336](end_span)[span_337](start_span)[span_337](end_span)[span_338](start_span)[span_338](end_span)75, \text{supp} \ge 5 | noul question testing universal entailment x[span_339](start_span)[span_339](end_span)[span_340](start_span)[span_340](end_span)[span_341](start_span)[span_341](end_span)\,p\,y \implies x\,q\,y |
| **Inverse Roles** | p \equiv q^- | AMIE+ hc(p^- | noul question testing |

| OWL 2 DL Axiom Family | Formal DL Syntax | Journal-Grounded Statistical Criterion (AMIE+ / CELOE) | clef-flash /v1/systemone AbA Verification Schema |
|---|---|---|---|
| **(owl:invers[span_342](start_span)[span_342](end_span)[span_343](start_span)[span_343](end_span)eOf)** | | \Rightarrow q) \ge 0.05, \text{conf}_{\text{PCA}}(p^- \Rightarrow q) \ge 0.70, \text{supp} \ge 5 | converse equivalence x\,p\,y \iff y\,q\,x |
| **Role Chains (owl:propertyChainAxiom)** | p_1 \c[span_346](start_span)[span_346](end_span)irc p_2 \sqsubseteq p_3 | AMIE+ 2-hop hc(p_1 \circ p_2 \Rightarrow p_3) \ge 0.05, \text{conf}_{\text{PCA}} \ge 0.80, \text{supp} \ge 5 | noul question verifying compositional soundness of the 2-hop Horn rule |
| **Property Characteristics (7 Flags)** | \text{Func}(p), \text{InvFunc}(p), \text{Trans}(p), \text{Sym}(p), \text{Asym}(p), \text{Ref[span_349](start_span)[span_349](end_span)l}(p), \text{Irrefl}(p) | AMIE+ functionality func(p), ifunc(p) and empirical satisfaction ratios \rho \in [0, [span_355](start_span)[span_355](end_span)[span_356](start_span)[span_356](end_span)1] with support \ge 5 | Single /v1/systemone request with 7 concurrent AbA noul questions per object property |
| **Local Class Restrictions (owl:Restriction)** | C \sqsubseteq \exists r.D and C [span_359](start_span)[span_359](end_span)[span_360](start_span)[span_360](end_span)[span_361](start_span)[span_361](end_span)\sqsubseteq \forall r.D | CELOE heuristic \phi_{\text{CELOE}}(C, B) = Q(B) + \lambda(Q(B)-Q(C)) - \beta\vert{}B\vert{} \ge 0.70 over E^+(C) and E^-(C) | Ontolearn verbalization + paired noul truth valuation and [span_362](start_span)[span_362](end_span)[span_363](start_span)[span_363](end_span)score necessity rating (\ge 1.80) |

## 4. Weighted MaxSAT Graph Repair, Subsumption Caching, and Formal OWL 2 DL Reasoner Debugging

In OWL 2 DL, combining owl:TransitiveProperty or owl:propertyChainAxiom with owl:FunctionalProperty, owl:InverseFunctionalProperty, owl:AsymmetricProperty, or owl:IrreflexiveProperty violates the Global Restrictions on Simple Roles, causing standard DL reasoners such as HermiT and Pellet to reject the ontology as syntactically ill-formed. Furthermore, as emphasized by Lehmann & Völker (*Perspectives on Ontology Learning*, IOS Press), asserting C_1 \sqcap C_2 \sqsubseteq \bot while p has rdfs:domain C_1 and a subclass A \sqsubseteq C_2 satisfies A \sqsubseteq \exists p.D forces A \sqsubseteq C_1 \sqcap C_2 \sqsubseteq \bot, making class A unsatisfiable (incoherent) even if A has zero ABox instances (DVC_POST == 0).

Pipeline v3.0 enforces two stages of formal repair. First, algebraic and graph-theoretic repair applies structural subsumption caching rules (C \equiv D \sqcap E \implies C \sqsubseteq D; C \equiv \exists r.C' \land D \equiv \exists r.\top \implies C \sqsubseteq D), resolves subsumption

cycles via confidence-weighted Minimum Feedback Arc Set reduction (w(u,v) = P_{\text{clef}}(u \sqsubseteq v) \cdot W_-(u,v) \cdot (\text{score}/3.0)), collapses mutual high-probability edges (\ge 0.85 in both directions) into owl:equivalentClass, and computes the Hasse diagram (G_{\text{reduced}}). Non-simple properties (Transitive or super-properties of propertyChainAxiom) are stripped of incompatible simple-role characteristics (Functional, InverseFunctional, Asymmetric, Irreflexiv[span_143](start_span)[span_143](end_span)e). Second, formal reasoner-in-the-loop debugging loads the induced TBox into an isolated owlready2.World() and executes sync_reasoner_h[span_371](start_span)[span_371](end_span)[span_372](start_span)[span_372](end_span)[span_373](start_span)[span_373](end_span)ermit() or sync_reasoner_pellet(). Any class returned in world.inconsistent_classes() (C \equiv \text{owl:Nothing}) triggers automated unsatisfiability pinpointing, iteratively retracting the lowest-weight disjointness or domain/range axiom along the ancestor path of C until zero inconsistent classes remain.

## 5. Self-Supervised Mask-and-Recover Benchmark, NeOn-GPT Lexical Alignment, Calibration Telemetry, and Multimodal Vision QA

To generate empirical results suitable for a top-tier journal submission directly from a single ABox input without human labels, Section 7 of v3.0 implements four automated evaluation suites:

1. **Self-Supervised Mask-and-Recover Benchmark (Ontolearn / OntoAxiom protocol):** Prior to concept induction, the pipeline creates a stratified hold-out split by masking 20\% of redundant explicit rdf:type assertions on multi-typed entities and holding out 15\% of closed 2-hop or inverse role edges; after full pipeline execution, it measures how accurately the induced TBox recovers the masked ground-truth assertions via Precision, Recall, and F1.
2. **NeOn-GPT Three-Pillar Structural & Jaro-Winkler (\ge 0.80) Lexical Alignment:** Following Ahmad et al. (*Semantic Web Journal*), the pipeline computes structural profile metrics alongside Jaro-Winkler lexical similarity between induced concepts/roles and masked or existing domain vocabulary terms (reporting the share of alignments achieving Jaro-Winkler \ge 0.80).
3. **Epistemic Calibration Telemetry:** Evaluates clef-flash's probabilistic outputs across M=10 uniform probability bins, computing the Expected Calibration Error (\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{N} |\text{acc}(B_m) - \text{conf}(B_m)|) and Brier Score (\frac{1}{N}\sum_{i=1}^N (p_i - y_i)^[span_385](start_span)[span_385](end_span)[span_386](start_span)[span_386](end_span)[span_387](start_span)[span_387](end_span)[span_388](start_span)[span_388](end_span)2) against the statistical baseline.
4. **Multimodal Topological Auditing:** Exploits clef-flash's native Qwen3VLProcessor joint image-text scoring in /v1/systemone by encoding the rendered Hasse taxonomic DAG figure (telemetry_diagnostics.png) as a base64 PNG and submitting it to /v1/systemone (images: [b64_png]) alongside structural metrics in state to obtain single-pass noul and score assessments of hierarchy balance and topological clarity.

# Complete Implementation Specification (.md) for the

# Antigravity AI Agent

The following Markdown specification is designed as a direct, drop-in replacement for universal_abox_to_tbox_agent_instructions.md. It instructs the Antigravity AI agent to generate a complete, self-contained abox2tbox.[span_125](start_span)[span_125](end_span)ipynb notebook that preserves the linear, zero-def cell-by-cell execution contract of v2.3 while integrating clef-flash via /v1/systemone and /api/chat, formal journal-grounded algorithms (CELOE, AMIE+, Ontolearn, OntoAxiom, NeOn-GPT, FCA), and self-supervised academic evaluation.

# Autonomous Agent Implementation Specification: Universal ABox-to-TBox Neuro-Symbolic Ontology Induction Pipeline (v3.0 — Clef-Flash & Journal-Grounded Edition)

## 1. Mission, Research Contributions & Target Deliverable Contract

### 1.1 Primary Objective

The autonomous agent shall generate and execute a self-contained, production-grade, end-to-end Jupyter Notebook (abox2tbox.ipynb) that induces a logically coherent, expressive OWL 2 DL Terminological Box (TBox.ttl) from an arbitrary Assertional Box (ABox.ttl) with zero human intervention.

### 1.2 Core Journal-Grounded Methodological Contributions Implemented by the Pipeline

The generated pipeline must implement five formal methodological contributions grounded in high-impact Semantic Web and Machine Learning journal literature:

1. **Dual-Endpoint Single-Model Neuro-Symbolic Mediation & OntoAxiom AbA Slot Isolation (*Semantic Web Journal*):** Uses Cloudflare's clef-flash (a 9B multimodal decision model based on Qwen3.5-9B with a non-autoregressive Joint Schema Head) across two Ollama endpoints: POST /v1/systemone for scoring multi-question schemas (noul, choice, score) in a single non-autoregressive forward pass (0 to 1 output token) with up to 32 isolated Axiom-by-Axiom (AbA) questions per request (achieving the AbA F1 advantage demonstrated by Bakker et al. in OntoAxiom without sequential generation overhead), and POST /api/chat strictly in Stage 2 to formulate NeOn-style **Competency Questions (CQs)**, candidate PascalCase class identifiers, and natural language glosses

for untyped entity clusters before routing those candidates back to /v1/systemone for discriminative selection.

2. **Open-World Soft Characteristic Clustering, Jaro-Winkler Lexical Deduplication & FCA Concept Lattice Intent Subsumption (*IEEE Intelligent Systems* & *Semantic Web Journal*):** Replaces brittle exact predicate hashing with Inverse-Document-Frequency (IDF) Weighted Jaccard Agglomerative Clustering over typed property signatures, applies Jaro-Winkler string similarity (\ge 0.80, following NeOn-GPT) to prevent near-duplicate class naming, and resolves the orphaned induced class defect via Formal Concept Analysis (FCA, Ganter & Wille) Intensional Signature Subsumption (\mathcal{P}_{\text{co[span_282](start_span)[span_282](end_span)[span_285](start_span)[span_285](end_span)re}}(C_2) \subseteq \mathcal{P}_{\text{core}}(C_1) \implies C_1 \sqsubseteq C_2) so newly discovered classes integrate seamlessly into the existing taxonomic DAG.
3. **AMIE+ Partial Completeness Assumption (\text{conf}_{\text{PCA}}) Role Mining & CELOE Downward Refinement Restrictions (*The VLDB Journal*, *JMLR*, & *Journal of Web Semantics*):** Expands ontological expressivity to OWL 2 DL (\mathcal{SROIQ}^{(D)}) by mining role hierarchies (rdfs:subPropertyOf), inverse properties (owl:inverseOf), and compositional role chains (owl:propertyChainAxiom) using AMIE+ Predicate Functionality (func(p), ifunc(p)), Head Coverage (hc \ge 0.05), and Orientation-Aware Partial Completeness Assumption confidence (\text{conf}_{\text{PCA}}) (Galárraga et al., *The VLDB Journal*). Induces local Description Logic class restrictions (owl:someValuesFrom \exists r.D and owl:allValuesFrom \forall r.D) using the CELOE complexity-penalized refinement heuristic \phi_{\text{CELOE}}(C, B) = Q(B) + \lambda(Q(B) - Q(C)) - \beta\vert{}B\vert{} over positive examples E^+(C) and superclass-derived negative examples E^-(C) = R(\text{Super}(C)) \setminus R(C) (Lehmann et al., *JMLR* & *Journal of Web Semantics*), paired with Ontolearn natural-language DL verbalization (*JMLR* 2025).
4. **Top-Down Sibling Disjointness Pruning, Structural Subsumption Caching & Reasoner-in-the-Loop Debugging (*IOS Press Studies on the Semantic Web*):** Eliminates quadratic disjointness explosion and heuristic fallback leaks by restricting disjointness candidates to taxonomic roots and Hasse siblings, caches structural subsumption tautologies, repairs cycles via Wilson-score-weighted Minimum Feedback Arc Set reduction, enforces OWL 2 DL simple-role restrictions across all 7 property characteristics, and verifies TBox class satisfiability (C \not\equiv \bot) via owlready2 (HermiT/Pellet) with automated unsatisfiability pinpointing.
5. **Self-Supervised Mask-and-Recover Benchmark, NeOn-GPT Three-Pillar Alignment, Calibration Telemetry & Multimodal Vision QA:** Executes a Self-Supervised Mask-and-Recover Evaluation (holding out 20% of multi-type assertions and 15% of closed relational edges) to report Precision, Recall, F1, Jaro-Winkler (\ge 0.80) lexical alignment, Expected Calibration Error (ECE), and Brier Score, culminating in a multimodal vision pass where the rendered Hasse DAG PNG is scored jointly with structural telemetry by clef-flash via /v1/systemone.

## 1.3 Strict Code Style & Operational Resilience Contract

The notebook must execute linearly from top to bottom across structured cells (Section 0 through Section 7) with zero multi-line def functions or custom Python class definitions, relying

exclusively on inline procedural loops, comprehensions, and single-line lambda helpers with explicit Python type annotations (dict[str, Any], list[tuple[str, ...]], Callable[..., Any]) on all state variables. To ensure uninterrupted execution across all hardware environments, the agent must implement a Three-Tier Decision Transport Layer:

| Transport Tier | Trigger Condition | Execution Mechanism & Model Target | Output Contract |
|---|---|---|---|
| **Tier 1: Native SystemOne HTTP** | /v1/systemone probe returns HTTP 200 | HTTP POST via urllib.request to {OLLAMA_BASE_URL}/v[span_93](start_span)[span_93](end_span)1/systemone with clef-flash:latest (or clef:latest[span_420](start_span)[span_420](end_span)[span_421](start_span)[span_421](end_span), kev:latest, nimble:latest, tev1:latest) | Native SystemOne JSON {"answers": ..., "usage": {"input_tokens": int, "output_tokens": int}} |
| **Tier 2: Chat JSON Emulation** | /v1/systemone returns HTTP 400/404/500 (e.g., Ollama v0.35.1 Issue #18769 or chat-only server) | Batched structured JSON schema call via ollama_client.chat using clef-flash:latest, qwen3.5:latest, or qwen2.5:latest | Emulated SystemOne {"answers": ...} with normalized negative Shannon entropy for confidence |
| **Tier 3: Statistical Fallback** | Ollama server unreachable or USE_LLM=False | Deterministic Wilson Score lower-bound, CELOE $\phi_{\text{CELOE}}$, and AMIE+ $\text{conf}_{\text{PCA}}$ estimators | Identical tuple schemas tagged with source = "statistical" |

All token telemetry recorded in PAYLOAD_TOKEN_AUDIT_LOG must be measured directly from live response metadata (usage.input_tokens / usage.output_tokens on /v1/systemone or pr[span_267](start_span)[span_267](end_span)ompt_eval_count / [span_424](start_span)[span_424](end_span)eval_count on /api/chat) and never fabricated.

# 2. Section-by-Section Notebook Architecture & Mathematical Specifications

## Section 0: Environment Setup, Configuration & Dual-Endpoint clef-flash Probe

Cell 0.0 installs rdflib, networkx, matplotlib, numpy, ollama, and owlready2 (%pip install -q rdflib networkx matplotlib numpy ollama owlready2). Cell 0.1 imports standard modules (collections,

typing, hashlib, itertools, json, math, os, random, re, sqlite3, time, base64, urllib.request, urllib.error, difflib) alongside matplotlib.pyplot, networkx, numpy, ollama, and rdflib, initializing seed_value: int = 42. Cell 0.2 defines config: dict[str, Any] and core mathematical lambda helpers according to the following parameter specification:

| Configuration Key | Default Value | Mathematical Role in Pipeline v3.0 (Journal Grounding) |
|---|---|---|
| "OLLAMA_BASE_URL" | "http://100.75.221.29:11434" | Primary Ollama server URL (with automatic fallback probe to "http://localhost:11434") |
| "MODEL_DECISION" | "clef-flash:latest" | Target model for /v1/sy[span_425](start_span)[span_425](end_span)[span_427](start_span)[span_427](end_span)stemone non-autoregressive AbA decision scoring |
| "MODEL_GENERATIVE" | "clef-flash:latest" | Target model for Stage 2 /api/chat NeOn-GPT Competency Question & class name synthesis |
| "SYSTEMONE_BATCH_SIZE" | 16 | Number of candidate axioms evaluated in isolated AbA question slots per /v1/systemone call (32 questions/call) |
| "MAX_DECISION_PAIRS" | 1000 | Maximum candidate pairs evaluated per stage (eliminating the v2.3 300-pair cap) |
| "TAU_MERGE_JACCARD" | 0.65 | Minimum IDF-weighted Jaccard similarity to merge untyped BLAKE2b micro-clusters |
| "TAU_INTENSIONAL_SUB" | 0.50 | Minimum FCA concept lattice intent inclusion ratio S_{\text{int}}(C_1 \sqsubseteq C_2) for induced classes |
| "TAU_SUB" / "TAU_SIG" | 0.15 / 0.35 | Minimum extensional conditional inclusion P(C_2 \mid C_1) and Jaccard similarity |
| "WILSON_Z" | 1.28 | Z-score for 80% Wilson Score lower bound W_-(k, n, z) on finite ABox counts |
| "AMIE_MIN_HC" | 0.05 | Minimum AMIE+ Head Coverage hc(\vec{B} \Rightarrow r(x, y)) for role rules (*The VLDB Journal*) |

| Configuration Key | Default Value | Mathematical Role in Pipeline v3.0 (Journal Grounding) |
|---|---|---|
| "AMIE_MIN_PCA_CONF" | 0.75 | Minimum AMIE+ Partial Completeness Assumption confidence \text{conf}_{\text{PCA}} (*The VLDB Journal*) |
| "CELOE_LAMBDA" / "CELOE_BETA" | 0.20 / 0.03 | Accuracy-gain weight \lambda and expression-length penalty \beta in \phi_{\text{CELOE}} (*JWS* / *JMLR*) |
| "JARO_WINKLER_THRESHOLD" | 0.80 | Minimum lexical similarity threshold for NeOn-GPT alignment and deduplication (*SWJ*) |
| "THETA_PROB" | 0.70 | Minimum clef-flash choice or noul probability required for axiom admission |
| "THETA_CONF" | 0.55 | Minimum hybrid confidence 0.5 \cdot W_- + 0.5 \cdot \text{confidence}_{\text{clef}} |
| "THETA_SCORE" | 1.80 | Minimum clef-flash expected ontological necessity score \mathbb{E}[L] \in [0, 3] |
| "THETA_NOUL" | 0.80 | Minimum noul probability for property characteristics, inverses, and role chains |
| "THETA_DOMAIN_RANGE" | 0.90 | Minimum closed-extension coverage required to assert global rdfs:domain or rdfs:range |
| "THETA_LOCAL_RESTRICTION" | 0.70 | Minimum CELOE heuristic score \phi_{\text{CELOE}}(C, B) to propose C \sqsubseteq \exists r.D or C \sqsubseteq \forall r.D |
| "HOLD_OUT_FRACTION" | 0.20 | Fraction of multi-type assertions masked for self-supervised F1 evaluation |

Cell 0.2 must define:

1. wilson_lower: Callable[[int, int, float], float] to compute: W_-(k, n, z) = \frac{\hat{p} + \frac{z^2}{2n} - z\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}} \quad \text{for } n > 0 \text{ else } 0.0
2. shannon_concentration: Callable[[list[float]], float] to compute 1.0 + \fr[span_3](start_span)[span_3](end_span)ac{\sum_i p_i \ln(\max(p_i, 10[span_306](start_span)[span_306](end_span)^{-12}))}{\ln(K)} for K > 1 options.
3. celoe_heuristic: Callable[[float, float, i[span_180](start_span)[span_180](end_span)[span_184](start_span)[span_184](end_sp

an)nt, float, float], float] implementing the CELOE objective (*Journal of Web Semantics*, Lehmann et al. 2011): \phi_{\text{CELOE}}(q_{\text{parent}}, q_{\text{ref}}, \text{length}, \lambda, \beta) = q_{\text{ref}} + \lambda \cdot (q_{[span_21](start_span)[span_21](end_span)\text{ref}} - q_{\text{parent}}) - \beta \cdot \text{length}

4. jaro_winkler_approx: Callable[[str, str], float] computing string similarity in [0, 1] with prefix boost (for NeOn-GPT lexical alignment at threshold \ge 0.80).

Cell 0.3 probes /api/tags and sends a live test request to {endpoint}/v1/systemone with {"model": active_decision_model, "state": {"probe": "readiness"}, "questions": {"ready": {"type": "noul", "instructions": "Is this a readiness check?"}}} to select Tier 1 (native_http), Tier 2 (chat_emulated), or Tier 3 (statistical).

## Section 1: ABox Parsing, Self-Supervised Masking & Soft Characteristic Clustering

Cell 1.1 parses ABOX_P[span_264](start_span)[span_264](end_span)ATH into rdflib.Graph() and populates the in-memory SQLite table triples(subject TEXT, predicate TEXT, object TEXT, is_literal INTEGER, datatype TEXT) with indices on subject, predicate, and object. After filtering out meta-vocabulary declarations (owl:Class, owl:ObjectProperty, owl:DatatypeProperty, owl:Ontology), Cell 1.1 constructs the self-supervised evaluation split by identifying all entities with \ge 2 explicit domain rdf:type assertions, deterministically sampling 20% (HOLD_OUT_FRACTION), deleting one redundant rdf:type assertion per sampled entity from SQLite while storing (entity, masked_class) in held_out_type_assertions, and holding out 15% of closed 2-hop or reciprocal relational edges in held_out_role_[span_278](start_span)[span_278](end_span)[span_280](start_span)[span_280](end_span)assertions.

Cell 1.2 extracts outgoing predicates, outgoing literal predicates, and incoming predicates for all untyped_subjects, forming initial 64-bit BLAKE2b micro-clusters (hash_to_entities). It then computes predicate Inverse Document Frequency \text{idf}(p) = \ln(1 + \[span_6](start_span)[span_6](end_span)vert{}I\vert{} / \text{df}(p)) across all instance subjects I and performs Agglomerative OWA Cluster Consolidation: any two micro-clusters K_a, K_b whose core predicate sets \mathcal{P}_a, \mathcal{P}_b (predicates present in \ge 50\% of cluster members) satisfy J_w(\mathcal{P}_a, \mathcal{P}_b) \[span_128](start_span)[span_128](end_span)ge \tau_{\text{merge}} (0.65) and share compatible incoming predicates are merged into a unified concept cluster.

## Section 2: Dominant Namespace Detection & NeOn-GPT Competency-Grounded Concept Synthesis

Cell 2.1 identifies detected_namespace by ranking non-standard URI namespaces across explicit classes, predicates, and subjects. Cell 2.2 iterates over each consolidated untyped cluster to synthesize open-vocabulary concepts via the Dual-Endpoint pattern grounded in the NeOn-GPT methodology (*Semantic Web Journal*, Ahmad et al.):

1. First, it extracts camelCase and snake_case token combinations as statistical fallback candidates.
2. Second, it calls active_generative_model (clef-flash) via /api/chat with a JSON schema requiring {"competency_question": str, "candidate_names": list[str], "ontological_gloss":

str}: the model must first formulate a NeOn **Competency Question (CQ)** capturing the cluster's domain semantics (e.g., *"Which domain records capture timestamped sensor readings and measurement units for an infrastructure asset?"*), then propose 3 singular PascalCase class identifiers and a 1-sentence ontological gloss grounded in the cluster's outgoing predicates, incoming predicates, and sample literal facts.

3. Third, it passes the Competency Question, cluster profile, and all candidate names (statistical + generative) into active_decision_model (clef-flash) via /v1/systemone with two concurrent questions: "best_name" (type: "choice", mapping each candidate identifier to "Candidate OWL 2 PascalCase class name answering the Competency Question") and "coherence" (type: "score", with 4 ordered criteria from "0: Accidental gro[span_439](start_span)[span_439](end_span)[span_440](start_span)[span_440](end_span)uping" to "3: Strict natural kind or institutional type").
4. Fourth, using jaro_winkler_approx, the winning choice is checked against existing_class_uris: if a pre-existing class shares Jaro-Winkler similarity \ge 0.92 AND identical core predicates, it is logged for equivalence alignment; otherwise it is sanitized, deduplicated against used_class_uris, materialized into SQLite as rdf:type assertions, and stored with its Competency Question (cq), glos[span_441](start_span)[span_441](end_span)[span_442](start_span)[span_442](end_span)s, and core predicate intent \mathcal{P}_{\text{core}}(C).

## Section 3: FCA Intensional Taxonomic Mining, AMIE+ PCA Role Mining & CELOE Class Expression Refinement

Cell 3.1 constructs class_ins[span_129](start_span)[span_129](end_span)tances, entity_direct_types, and class_core_predicates: dict[str, set[str]] (predicates occurring on \ge 50\% of instances of each class C). For pairs (C_1, C_2) with shared instances (overlap_count > 0), it computes conditional inclusions P(C_2 \mid C_1) and P(C_1 \mid C_2), Wilson lower bounds W_-(\vert{}C_1 \cap C_2\vert{}, \vert{}C_1\vert{}, z) and W_-(\vert{}C_1 \cap C_2\vert{}, \vert{}C_2\vert{}, z), and Jaccard similarity, admitting pairs with \max(P(C_2 \mid C_1), P(C_1 \mid C_2)) \ge \tau_{\text{sub}} or \text{Jaccard} \ge \tau_{\text{sig}} tagged with mining_mode = "extensional". To integrate newly induced classes that have zero instance overlap with existing classes, Cell 3.1 implements Formal Concept Analysis (FCA, Ganter & Wille) Intensional Signature Inclusion:

$$S_{\text{int}}(C_1 \sqsubseteq C_2) = \frac{\vert{}\mathcal{P}_{\text{core}}(C_1) \cap \mathcal{P}_{\text{core}}(C_2)\vert{}}{\vert{}\mathcal{P}_{\text{core}}(C_2)\vert{}}$$

for pairs where \vert{}C_1 \cap C_2\vert{} = 0 and at least one class belongs to induced_class_uris. Pairs satisfying \max(S_{\text{int}}(C_1 \sqsubseteq C_2), S_{\text{int}}(C_2 \sqsubseteq C_1)) \ge \tau_{\text{intensional_sub}} (0.50) with at least 2 shared core predicates are appended to candidate_taxonomic_pairs tagged with mining_mode = "intensional_fca".

Cell 3.2 profiles all domain properties and mines expressive DL candidates using AMIE+ (*The VLDB Journal*) and CELOE / Ontolearn (*JMLR* & *Journal of Web Semantics*):

1. **AMIE+ Functionality & 7 OWL 2 Characteristics:** For each property p, Cell 3.2 computes func_score = func(p) = \vert{}S_p\vert{} / \vert{}E_p\vert{}, ifunc_score = ifunc(p) = \vert{}O_p\vert{} / \vert{}E_p\vert{}, and empirical satisfaction ratios for all 7 OWL 2 characteristics: \rho_{\text{func}}, \rho_{\text{invfunc}}, \rho_{\text{trans}},

\rho_{\text{sym}}, \rho_{\text{asym}} = 1 - \rho_{\text{sym}}, \rho_{\text{refl}} = \frac{\vert{}\{(s,o) \in E_p \mid s = o\}\vert{}}{\vert{}S_p\vert{}}, and \rho_{\text{irrefl}} = 1 - \rho_{\text{refl}}.

2. **AMIE+ Orientation-Aware PCA Role Mining (rdfs:subPropertyOf, owl:inverseOf, owl:propertyChainAxiom):**
   - For ordered object property pairs (p, q), it computes support \text{supp}(p \Rightarrow q) = \vert{}E_p \cap E_q\vert{}, Head Coverage hc(p \Rightarrow q) = \text{supp}(p \Rightarrow q) / \ve[span_351](start_span)[span_351](end_span)rt{}E_q\vert{}, and orientation-aware PCA confidence: \text{conf}_{\text{PCA}}(p \R[span_147](start_span)[span_147](end_span)[span_151](start_span)[span_151](end_span)ightarrow q) = \frac{\vert{}E_p \cap E_q\vert{}}{\vert{}\{(x, y) \in E_p \mid x \in S_q\}\vert{}} \text{ if } func(q) \ge ifunc(q) \text{ else } \frac{\vert{}E_p \cap E_q\vert{}}{\vert{}\{(x, y) \in E_p \mid y \in O_q\}\vert{}} admitting candidate_subproperties when hc \ge 0.05, \text{conf}_{\text{PCA}} \ge 0.75, and \text{supp} \ge 5.
   - For inverse role candidates (p, q), it evaluates R: p(y, x) \Rightarrow q(x, y) with support \vert{}\{(x, y) \in E_q \mid (y, x) \in E_p\}\vert{}, hc \ge 0.05, and [span_443](start_span)[span_443](end_span)[span_444](start_span)[span_444](end_span)\text{conf}_{\text{PCA}}(p^- \Rightarrow q) \ge 0.70.
   - For object property triples (p_1, p_2, p_3), it evaluates 2-hop Horn rules p_1(x, z) \land p_2(z, y) \Rightarrow p_3(x, y) using AMIE+ hc \ge 0.05, \text{conf}_{\text{PCA}} \ge 0.80, and support \ge 5 (candidate_property_chains).
3. **CELOE Downward Refinement & Ontolearn Verbalization for Local Class Restrictions (C \sqsubseteq \exists r.D, C \sqsubseteq \forall r.D):**
   - Following Lehmann et al. (*Journal of Web Semantics*, 2011), for each class C (\vert{}C\vert{} \ge 5), positive examples are E^+(C) = \text{class_instances}[C] and negative examples are E^-(C) = \{e \in I \setminus E^+(C) \mid \text{entity shares a candidate superclass or [span_205](start_span)[span_205](end_span)[span_208](start_span)[span_208](end_span)predicate domain with } C\} (or I \setminus E^+(C) if empty).
   - For each object property r and target filler class D occurring on instances of C, Cell 3.2 evaluates the downward refinements B_{\exists} = \exists r.D (length = 3) and B_{\forall} = \forall r.D (length = 3). It computes recall/coverage on E^+(C), precision separating E^+(C) from E^-(C), F1 quality Q(B), and the CELOE heuristic \phi_{\text{CELOE}}(C, B) = Q(B) + \lambda(Q(B) - Q_0) - \beta \cdot 3.
   - Whenever \phi_{\text{CELOE}}(C, B) \ge \text{THETA_LOCAL_RESTRICTION} (0.70) and positive coverage \ge 0.85 (for \exists r.D) or filler purity \ge 0.95 across \ge 5 edges (for \fo[span_446](start_span)[span_446](end_span)[span_448](start_span)[span_448](end_span)rall r.D), Cell 3.2 generates an Ontolearn-style natural language verbalization (*JMLR* 2025, e.g., "Every instance of {C} is linked via {r} to at least one instance of {D}" or "Every instance of {C} linked via {r} has only fillers of type {D}") and appends the candidate to candidate_local_restrictions.

## Section 4: Batched Clef-Flash /v1/systemone Axiom-by-Axiom (AbA)

## Adjudication

Following the OntoAxiom benchmark finding (*Semantic Web Journal*, Bakker et al.) that Axiom-by-Axiom (AbA) evaluation prevents cross-axiom interference and maximizes F1, Cell 4.1 partitions candidate_taxonomic_pairs into batches of SYSTEMONE_BATCH_SIZ[span_450](start_span)[span_450](end_span)[span_451](start_span)[span_451](end_span)[span_452](start_span)[span_452](end_span)E (16 pairs = 32 isolated AbA question slots per request). For each batch, the state object contains each pair's class labels, instance counts, intersection size, conditional probabilities, Wilson lower bounds, Jaccard similarity, Jaro-Winkler lexical similarity, shared core predicates, and mining_mode. Each pair i in the batch is evaluated in its own isolated /v1/systemone AbA slot via two concurrent questions: "rel_{i}" (type: "choice", with options "c1_sub_c2", "c2_sub_c1", "equivalent", and "none[span_319](start_span)[span_319](end_span)[span_321](start_span)[span_321](end_span)[span_323](start_span)[span_323](end_span)") and "[span_426](start_span)[span_426](end_span)[span_428](start_span)[span_428](end_span)nec_{i}" (type: "score"[span_182](start_span)[span_182](end_span)[span_186](start_span)[span_186](end_span), with 4 ordered levels from "0: Accidental dataset artifact" to "3: Strict analytical or taxonomic necessity"). From the single-pass response, Cell 4.1 extracts decision = answers[f"rel_{i}"]["choice"], prob = answers[f"rel_{i}"]["probabilities"][decision], model_conf = answers[f"rel_{i}"]["confidence"], and score = answers[f"nec_{i}"]["score"], computing hybrid confidence conf = 0.5 * wilson_ev + 0.5 * model_conf. Axioms are admitted when decision != "none", prob >= THETA_PROB, conf >= THETA_CONF, and score >= THETA_SCORE (requiring prob >= 0.80 and score >= 2.10 for intensional_fca bridges).

Cell 4.2 constructs a preliminary Hasse DAG G_prelim from adjudicated_subsumptions and eliminates the v2.3 608-pair disjointness fallback leak via Top-Down Sibling and Root Filtering. Because disjointness is inherited downward by subclasses ($A \sqcap B \sqsubseteq \bot \land C \sqsubseteq A \implies C \sqcap B \sqsubseteq \bot$), Cell 4.2 restricts candidate_disjoint_pairs strictly to pairs ($C_1$, $C_2$) with $\le \epsilon_{\text{disj}}$ overlap and support $\ge 5$ that either share a direct parent in G_prelim (Hasse siblings) or both have $\text{in-degree} = 0$ in G_prelim (taxonomic roots). These structural pairs are batched (16 pairs per /v1/systemone call) with "disj_{i}" (type: "noul", probability of strict mutual exclusion) and "disj_nec_{i}" (type: "score", 4 incompatibility levels), admitting only pairs that satisfy disj_noul >= THETA_PROB, disj_score >= THETA_SCORE, and Wilson non-overlap bound $\ge \text{THETA_CONF}$.

Cell 4.3 adjudicates all property characteristics, AMIE+ role hierarchies, inverse properties, property chains, and CELOE local class restrictions via /v1/systemone. Each property is evaluated in a single /v1/systemone request containing up to 7 AbA noul questions (is_functional, is_inverse_functional, is_transitive, is_symmetric, is_asymmetric, is_reflexive, is_irreflexive). Constraining characteristics (Functional, InverseFunctional, Asymmetric, Irreflexive) are gated by $\min(\text{noul}_{\text{clef}}, \rho_{\text{empirical}})$ so the TBox never contradicts ABox data, whereas open-world generative characteristics (Transitive, Symmetric, Reflexive) are gated by $\text{noul}_{\text{clef}} \ge \text{THETA_NOUL}$ given support $\ge 5$. Candidate AMIE+ sub-properties, inverse roles, and role chains (with their $\text{conf}_{\text{PCA}}$ and hc statistics in state) and CELOE local restrictions ($C \sqsubseteq \exists r.D$, $C$ [span_265](start_span)[span_265](end_span)$\sqsubseteq \forall r.D$, with their $\phi_{\text{CELOE}}$ score and Ontolearn verbalization in state) are batched through

/v1/s[span_352](start_span)[span_352](end_span)ystemone noul and score questions and admitted when noul >= THETA_NOUL and score >= THETA_SCORE.

## Section 5: Weighted MaxSAT Graph Repair, Subsumption Caching, Simple-Role Enforcement & Reasoner Debugging

Cell 5.1 applies structural subsumption caching heuristics (C \equiv D \sqcap E \implies C \sqsubseteq D; C \equiv \exists r.C' \land D \equiv \exists r.\top \implies C \sqsubseteq D), constructs G_sub with edge weights w(u,v) = \text{prob} \cdot \text{conf} \cdot (\text{score} / 3.0), collapses mutual high-weight 2-cycles (\ge 0.65 in both directions) into final_equivalences, iteratively removes the minimum-weight edge from any remaining directed cycle (nx.find_cycle(G_sub)), and computes the minimal Hasse reduction G_reduced = nx.transitive_reduction(G_sub).
Cell 5.2 computes ancestor closures over G_reduced and final_equivalences, removing any disjoint pair (C_1, C_2) that shares a common subclass (D \sqsubseteq^* C_1 \land D \sqsubseteq^* C_2) or a common ABox instance (DVC_POST == 0), and prunes redundant descendant disjointness assertions whenever an ancestor disjointness axiom is already present. It then enforces OWL 2 DL syntactic restrictions on simple roles: any property declared Transitive or serving as the super-property of an owl:propertyChainAxiom is stripped of Asymmetric, Irreflexive, Functional, and InverseFunctional flags, and pairwise mutexes (Symmetric vs. Asymmetric, Reflexive vs. Irreflexive) are resolved by retaining the higher probability characteristic.
Cell 5.3 assigns global rdfs:domain (\e[span_463](start_span)[span_463](end_span)[span_464](start_span)[span_464](end_span)xists r.\top \sqsubseteq C) and rdfs:range (\top \sqsubseteq \forall r.C) to the most specific class covering \ge \text{THETA_DOMAIN_RANGE} (0.90) of closed subjects or objects (suppressing overly restrictive global domain/range when coverage < 0.90 in favor of CELOE local owl:Restriction axioms). Following Lehmann & Völker (*Perspectives on Ontology Learning*, IOS Press), Cell 5.3 then couples ontology learning with formal DL inconsistency debugging: it loads the induced TBox into an isolated owlready2.World() and executes owlready2.sync_reasoner_hermit(world, debug=0) (falling back gracefully to sync_reasoner_pellet or symbolic graph-coherence verification if a local JVM is unavailable), verifying that list(world.inconsistent_classes()) is empty and iteratively retracting any offending lowest-weight disjointness or domain/range axiom if an unsatisfiable class (C \equiv \bot) is detected.

## Section 6: OWL 2 DL TBox Materialization with W3C PROV-O, Ontolearn Verbalizations & SystemOne Provenance

Cell 6.1 constructs g_tbox: rdflib.Graph containing the owl:Ontology header, owl:Class declarations with rdfs:label, NeOn-GPT competencyQuestion annotations, and rdfs:comment glosses, Hasse rdfs:subClassOf edges reified (when EXPORT_STRICT_OWL2_DL=False) with decisionProbability, confidenceScore, ontologicalNecessityScore, aboxSupportCount, miningMode, and prov:wasAttributedTo, alongside owl:equi[span_447](start_span)[span_447](end_span)[span_449](start_span)[span_449](end_span)valentClass, pruned owl:disjointWith, blank-node owl:Restriction axioms (owl:someValuesFrom and owl:allValuesFrom annotated with their Ontolearn natural-language

verbalization and \phi_{\text{CELOE}} score), rdfs:subPropertyOf and owl:inverseOf (annotated with AMIE+ \text{conf}_{\text{PCA}} and hc), owl:propertyChainAxiom RDF collections, and the 7 OWL 2 property characteristic types. Cell 6.2 serializes g_tbox to TBOX_PATH in Turtle format and executes a strict round-trip parse check (g_verification).

### Section 7: Self-Supervised Benchmark, NeOn-GPT Alignment, Epistemic Calibration (ECE/Brier) & Multimodal Vision QA

Cell 7.1 computes OntoQA and NeOn-GPT structural profile metrics (RelationshipRichness, InheritanceRichness, ClassCoverageRatio, AxiomExpressivityInd[span_354](start_span)[span_354](end_span)ex[span_273](start_span)[span_273](end_span), DisjointnessDensityRatio, and JaroWinklerLexicalAlignmentRate at threshold \ge 0.80) and evaluates recovery of held_out_type_assertions and held_out_role_assertions using the induced TBox entailments (domain/range typing, CELOE existential/universal restrictions, characteristic cluster matching, subclass closure, and AMIE+ role inverse/subproperty/chain completion), reporting Hold-Out Precision, Recall, and F1. Cell 7.2 evaluates epistemic calibration across M=10 probability bins to report Expected Calibration Error (ECE), Brier Score, total /v1/systemone calls, and measured input/output token counts. Cell 7.3 renders a 3-panel diagnostic figure (PLOT_PATH) showing the 10-bin Reliability Calibration Curve, Axiom Expressivity Breakdown, and Top-20 Hasse Taxonomic DAG (distinguishing existing and newly integrated induced classes), then base64-encodes the saved PNG and submits it to clef-flash via /v1/systemone (images: [b64_png], state: ontoqa_metrics) with "dag_well_formed" (type: "noul") and "topology_quality" (type: "score") to output a multimodal topological quality audit.

# Conclusions and Research Positioning

Upgrading the agent specification from v2.3 to v3.0 directly addresses every structural and epistemic limitation identified in abox2tbox.ipynb while anchoring every stage of the pipeline in high-impact peer-reviewed journal methodologies. By replacing autoregressive chat JSON generation (qwen2.5:latest, which consumed 8,349 completion tokens and forced a 300-pair truncation cap that leaked 608 uncalibrated disjointness axioms) with clef-flash's non-autoregressive Joint Schema Head via /v1/systemone, each multi-question adjudication batch executes isolated Axiom-by-Axiom (OntoAxiom, *Semantic Web Journal*) evaluations in a single forward pass (0–1 output tokens) while yielding calibrated softmax probabilities (probabilities), distributional sharpness (confidence), and probability-weighted necessity expectations (score).
Furthermore, pairing /v1/s[span_17](start_span)[span_17](end_span)ystemone discriminative scoring with Formal Concept Analysis Intensional Signature Subsumption (Ganter & Wille) resolves the orphaned-induced-class failure mode, while AMIE+ Orientation-Aware Partial Completeness Assumption (\text{conf}_{\text{PCA}}) role mining (*The VLDB Journal*), CELOE complexity-penalized downward refinement (\phi_{\text{CELOE}}) and Ontolearn DL verbalization (*Journal of Machine Learning Research* and *Journal of Web Semantics*), NeOn-GPT Competency Question grounding and Jaro-Winkler (\ge 0.80) lexical evaluation (*Semantic Web Journal*), and HermiT/Pellet tableau debugging (*Perspectives on Ontology Learning*, IOS Press) elevate the pipeline from a descriptive RDFS script to a rigorous,

self-evaluating neuro-symbolic research framework.

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